

Chapter 9

Multi-Criteria Decision-Making: Outranking-Type Methods

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Abstract

This chapter describes outranking-type MCDM methods, which evaluate alternatives based on pairwise comparisons and outranking relationships. Two widely used outranking method families (ELECTRE and PROMETHEE) are presented, with a focus on their most popular variants: ELECTRE III and PROMETHEE II. This chapter explains the theoretical foundations, algorithmic steps, and implementation details for each of these methods, supported by illustrative numerical examples using a common alternatives-criteria matrix (ACM). Key concepts such as the indifference, preference, and veto thresholds in ELECTRE methods, and the various preference functions in PROMETHEE methods, are explained. The advantages and limitations are also discussed. After learning the contents of this chapter, readers can (1) describe the principles and steps of outranking-type methods, and (2) apply them to MCDM problems.

9.1 Overview

Multi-criteria decision making (MCDM) provides a robust framework for identifying the most suitable alternative(s) from a set defined by an alternatives-criteria matrix (ACM) (Nabavi et al., 2024). This chapter focuses on a specific class of MCDM techniques known as outranking-type methods, which evaluate alternatives through pairwise comparisons across multiple criteria. These methods establish an outranking relationship by determining the degree to which one alternative is preferred over another.

This chapter explores 2 prominent families of outranking-type MCDM methods, namely, ELECTRE (Elimination et Choix Traduisant la Réalité; translated to: Elimination and Choice Expressing the Reality) and PROMETHEE (Preference Ranking Organization Method for Enrichment Evaluations). The most popular variant from each of these families is detailed and

illustrated using a simple numerical example, to provide an intuitive understanding of their underlying mechanisms. The following sections in this chapter present these methods. Readers need not study them sequentially and can read either of them.

Section 9.2 Elimination et Choix Traduisant la Realité (ELECTRE) Methods

Section 9.3 Preference Ranking Organization Method for Enrichment Evaluations (PROMETHEE) Methods

Section 9.4 Summary

Our MS Excel program, EMCDM, described in Chapter 10, includes two variants of ELECTRE.

The learning outcomes of this chapter are: (1) Describe the principles and algorithms of outranking-type MCDM methods covered, namely, ELCTRE and PROMETHEE; and (2) Apply any of these methods to ACM of an application.

Unless otherwise stated, the notation used throughout this chapter is as follows:

- $i \in \{1, 2, \dots, m\}$ and $j \in \{1, 2, \dots, n\}$, where m is the number of rows (i.e., alternatives) and n is the number of columns (i.e., criteria) in the ACM.
- f_{ij} is the value of the j^{th} criterion for the i^{th} alternative in the original ACM.
- F_{ij} is the value of f_{ij} after normalization/transformation.
- w_j is the weight assigned to the j^{th} criterion, and $\sum_{j=1}^n w_j = 1$.

Additionally, the ACM presented in Table 9.1 is employed to walk through the steps of ELECTRE and PROMETHEE methods covered in this chapter. This ACM is made up of 3 alternatives (i.e., $m = 3$ and $i \in \{1, 2, 3\}$) and 4 criteria (i.e., $n = 4$ and $j \in \{1, 2, 3, 4\}$). Out of the 4 criteria (C1, C2, C3, and C4), C1, C2, and C3 are benefit criteria to be maximized whereas C4 is a cost criterion to be minimized. For illustration, the weights for the criteria are assumed to be: $w_1 = 0.30$, $w_2 = 0.25$, $w_3 = 0.20$, and $w_4 = 0.25$. All the 3 alternatives (A1, A2, and A3) are non-dominated, which means that, among them, the improvement in one criterion will inevitably result in degradation in at least one other criterion (Wang et al., 2024).

Table 9.1: ACM employed for illustration of ELECTRE and PROMETHEE methods (C1, C2, and C3 for maximization; C4 for minimization).

Alternatives	C1 (Max)	C2 (Max)	C3 (Max)	C4 (Min)
A1	0.78	117	6.2	35
A2	0.65	150	5.8	40
A3	0.83	110	4.9	30
Weight	0.30	0.25	0.20	0.25

9.2 Elimination et Choix Traduisant la Realité (ELECTRE) Methods

The ELECTRE family of methods, originally proposed by Roy (1968), is a popular group of outranking techniques in MCDM, especially when dealing with complex problems involving conflicting criteria. These methods are designed to compare alternatives by constructing outranking relations through a series of pairwise comparisons across all criteria (Wang & Rangaiah, 2017), rather than relying on reference-based or aggregation-based scoring strategies.

ELECTRE family has been developed and refined over several decades, with several variants such as ELECTRE I, II, III, IV, IS, and TRI (Figueira et al., 2010; Govindan & Jepsen, 2016). It started with the introduction of ELECTRE I (Roy, 1968). Then, each subsequent variant addresses specific limitations or adapting to different decision-making contexts. ELECTRE I is the foundational model, developed for binary outranking decisions in choice problems, but it lacks the ability to provide rankings among alternatives (Govindan & Jepsen, 2016). Its improved version, ELECTRE II (Roy & Bertier, 1973) includes a mechanism for complete and partial ranking, making it more suitable for decision scenarios requiring prioritization (Lima & Soares, 2011).

Pseudo-criteria and credibility indices are introduced in ELECTRE III (Roy, 1978), to enable handling of more complicated information and allowing for more nuanced preference modeling in complex real-world applications (Figueira et al., 2010). ELECTRE IV (Roy & Hugonnard, 1982) extends these capabilities to settings with no available thresholds or preferences, making it applicable when limited quantitative information is present (Hokkanen & Salminen, 1997). ELECTRE IS (Roy & Bouyssou, 1993), an improved variant of ELECTRE I, is developed to simplify parameter elicitation (e.g., indifference thresholds) and improving the robustness of the conclusions (Park et al., 2005). ELECTRE TRI (Yu, 1992) introduces a sorting approach rather than ranking or choosing; it is designed to classify alternatives into predefined categories based on performance profiles, which made it ideal for applications like credit scoring or risk classification (Mousseau & Slowinski, 1998).

Among these variants, ELECTRE III stands out as the most widely used and recognized one for ranking alternatives (Figueira et al., 2010), which introduces the indifference, preference, and veto thresholds in its algorithm, allowing more nuanced comparisons between alternatives (Liao et al., 2020). The steps in ELECTRE III method are outlined below.

Step 1. Transform all the original criteria to maximization type:

$$F_{ij} = \begin{cases} f_{ij} & \text{if criterion } j \text{ is for maximization} \\ -f_{ij} & \text{if criterion } j \text{ is for minimization} \end{cases} \quad (9.1)$$

This transformation ensures that a higher F_{ij} always indicates a better performance.

Numerical Calculations (using ACM from Table 9.1):

For instance, for $i = 1$ and $j = 4$, since C4 is a minimization criterion:

$$F_{14} = -f_{14} = -35$$

Similar transformations are performed for all other values in the ACM across the $m = 3$ alternatives and $n = 4$ criteria. The complete results, forming the transformed ACM, are shown in Table 9.2.

Table 9.2: Transformed ACM for ELECTRE III walkthrough.

Alternatives	C1	C2	C3	C4
A1	0.78	117	6.2	-35
A2	0.65	150	5.8	-40
A3	0.83	110	4.9	-30

Step 2. Construct the concordance matrix by pairwise comparison; this matrix has m rows and m columns, corresponding to m alternatives. This construction is as follows.

Let a and b represent any two alternatives chosen for pairwise comparison. The indifference threshold (Q_j) defines the range of variations for criterion j within which the decision-maker perceives no significant difference between the two alternatives. In other words, it represents the interval where the performance of two alternatives on a given criterion is considered indistinguishable. When the difference in performance between two alternatives for a specific criterion exceeds the preference threshold (P_j), a clear preference is assigned to the alternative with the better value (i.e., higher F_{ij}). The veto threshold (V_j) is used to disqualify an alternative relative to another when the difference in performance for criterion j is too large to be deemed acceptable.

The Q_j , P_j , and V_j (all non-negative) can be given by decision-makers based on the context of the application. So, these inputs will be subjective. Rogers & Bruen (1998) stated that these thresholds can be determined as a proportion of the range of criterion values. In the numerical example presented here, as well as in our EMCDM program (described in Chapter 10), the thresholds: Q_j , P_j , and V_j are respectively set to 10%, 20%, and 80% of the range of criterion j (Lee & Rangaiah, 2009). These can be adjusted as and when necessary.

Each element of the concordance matrix, i.e., $C(a, b)$, is computed using:

$$C(a, b) = \sum_{j=1}^n w_j C_j(a, b), \quad a, b \in \{1, 2, \dots, m\} \quad (9.2a)$$

$$\text{Here, } C_j(a, b) = \begin{cases} 1 & \text{if } F_j(b) - F_j(a) \leq Q_j \\ 0 & \text{if } F_j(b) - F_j(a) > P_j \\ \frac{P_j - [F_j(b) - F_j(a)]}{P_j - Q_j} & \text{if } Q_j < F_j(b) - F_j(a) \leq P_j \end{cases} \quad (9.2b)$$

Essentially, $C_j(a, b)$ represents the degree of support from criterion j for the assertion that alternative a is at least as good as alternative b under this criterion. Here, $0 \leq C_j(a, b) \leq 1$. A value of $C_j(a, b) = 1$ means that criterion j fully supports this assertion, indicating that a performs as good as or better than b . Conversely, $C_j(a, b) = 0$ signifies that the performance gap exceeds the preference threshold P_j , meaning that a clearly is worse than b on criterion j . For the other $C_j(a, b)$ values between 0 and 1, the difference lies between Q_j and P_j , representing a partial support: values closer to 1 suggest near indifference between a and b , while values closer to 0 imply that a is much less favorable than b .

A special case arises when an alternative is compared with itself (i.e., $a = b$); in this situation, $C_j(a, b) = 1$ according to Eq. (9.2b). Next, $C(a, b)$ aggregates these degrees of support across all criteria, multiplied by their weights, to express the overall strength of evidence that a is at least as good as b .

Numerical Calculations:

As stated above, the thresholds: Q_j , P_j , and V_j for each criterion j are assumed to be, respectively, 10%, 20% and 80% of the range of the criterion. For instance, for $j = 2$, the thresholds of criterion C2 (whose maximum and minimum are respectively 150 and 110) are calculated as follows:

$$Q_2 = 0.1 \times (150 - 110) = 4$$

$$P_2 = 0.2 \times (150 - 110) = 8$$

$$V_2 = 0.8 \times (150 - 110) = 32$$

Comparing the two alternatives of $a = 3$ and $b = 1$ with respect to C2, $F_2(1) - F_2(3) = 117 - 110 = 7$. Since this difference is between $Q_2 = 4$ and $P_2 = 8$,

$$C_2(3, 1) = \frac{8 - [117 - 110]}{8 - 4} = 0.25$$

It should be noted that $C_j(a, b)$ need not be the same as $C_j(b, a)$ in the concordance matrix, according to Eq. (9.2b). For example, $C_2(1, 3) = 1$ as $F_2(3) - F_2(1) = 110 - 117 = -7$, which is less than $Q_2 = 4$.

Similarly, pairwise comparisons between these two alternatives are performed across the remaining 3 criteria; the computed values are: $C_1(3, 1) = 1$, $C_3(3, 1) = 0$, and $C_4(3, 1) = 1$. This is followed by the computation of the weighted aggregate, $C(3, 1) = \sum_{j=1}^n w_j C_j(3, 1) =$

0.6125 using the assigned weights in Table 9.1. The complete concordance matrix for all 3 alternatives considering all 4 criteria is displayed in Table 9.3. Note that, according to Eq. (9.2b), all diagonal elements are equal to 1 (representing an alternative compared with itself).

Table 9.3: Concordance matrix for ELECTRE III walkthrough.

$C(a, b)$	A1	A2	A3
A1	1	0.75	0.45
A2	0.25	1	0.45
A3	0.6125	0.55	1

Step 3. Construct the discordance matrix under each criterion, that is, $D_j(a, b)$.

$$D_j(a, b) = \begin{cases} 1 & \text{if } F_j(b) - F_j(a) > V_j \\ 0 & \text{if } F_j(b) - F_j(a) \leq P_j \\ \frac{F_j(b) - F_j(a) - P_j}{V_j - P_j} & \text{if } P_j < F_j(b) - F_j(a) \leq V_j \end{cases} \quad (9.3)$$

Here, $0 \leq D_j(a, b) \leq 1$. As the counterpart of $C_j(a, b)$, $D_j(a, b)$ reflects the degree to which, for a criterion j , the performance of alternative a is significantly worse than that of alternative b . $D_j(a, b) = 1$ means the difference surpasses the veto threshold V_j , signaling strong evidence from criterion j that a is significantly worse than b on this criterion. At the other extreme, $D_j(a, b) = 0$ means criterion j provides no indication that a is significantly worse than b .

When an alternative is compared to itself (i.e., $a = b$), then $D_j(a, b) = 0$. There will be n discordance matrices, each corresponding to one of n criteria.

Numerical Calculations:

For instance, for $j = 2$, comparing the two alternatives of $a = 3$ and $b = 1$: since $F_j(b) - F_j(a) = 117 - 110 = 7 \leq P_j = 8$ in this case, we get $D_2(3, 1) = 0$. Likewise, pairwise comparisons of the other alternatives under the 2nd criterion are conducted, and the resulting discordance matrix, $D_2(a, b)$, is presented in Table 9.4. Also, this table shows discordance matrices for the remaining 3 criteria, derived in the same manner.

Table 9.4: Discordance matrices for ELECTRE III walkthrough.

$D_1(a, b)$	A1	A2	A3
A1	0	0	0.1296
A2	0.8704	0	1
A3	0	0	0

$D_2(a, b)$	A1	A2	A3
A1	0	1	0
A2	0	0	0
A3	0	1	0
$D_3(a, b)$	A1	A2	A3
A1	0	0	0
A2	0.1795	0	0
A3	1	0.8205	0
$D_4(a, b)$	A1	A2	A3
A1	0	0	0.5
A2	0.5	0	1
A3	0	0	0

Step 4. Construct the credibility matrix $S(a, b)$ of size $m \times m$ as follows.

$$S(a, b) = \begin{cases} C(a, b) & \text{if } D_j(a, b) \leq C(a, b) \forall j \\ C(a, b) \cdot \prod_{j \in J} \frac{1 - D_j(a, b)}{1 - C(a, b)} & \text{otherwise} \end{cases} \quad (9.4)$$

Here, for the product term, the set, $J = \{j \mid D_j(a, b) > C(a, b)\}$. The quantity, $S(a, b)$ expresses the overall credibility or strength of the assertion that alternative a outranks alternative b considering both the concordance and discordance assessments.

Numerical Calculations:

For instance, consider the two alternatives: $a = 3$ and $b = 1$. The condition of $D_j(a, b) \leq C(a, b) \forall j$ is not satisfied because $D_3(3, 1) = 1$ (Table 9.4) and $C(3, 1) = 0.6125$ (Table 9.3). Consequently, $J = \{3\}$ and so

$$S(3, 1) = C(3, 1) \cdot \prod_{j \in J} \frac{1 - D_j(3, 1)}{1 - C(3, 1)} = 0.6125 \cdot \left(\frac{1 - 1}{1 - 0.6125} \right) = 0$$

Consider two other alternatives: $a = 3$ and $b = 3$. The condition of $D_j(a, b) \leq C(a, b) \forall j$ is satisfied because $D_1(3, 3) = D_2(3, 3) = D_3(3, 3) = D_4(3, 3) = 0.0$ (Table 9.4) and $C(3, 3) = 1.0$ (Table 9.3). Consequently, $S(3, 3) = C(3, 3) = 1.0$. Similarly, remaining elements of the credibility matrix are computed. The complete credibility matrix is presented in Table 9.5.

Table 9.5: Credibility matrix for ELECTRE III walkthrough.

$S(a, b)$	A1	A2	A3
A1	1	0	0.4091

A2	0.0288	1	0
A3	0	0	1

Step 5. Finally, calculate the outranking score (R_a) for each alternative a .

$$R_a = \sum_{b=1}^m S(a, b) - \sum_{b=1}^m S(b, a) \quad (9.5)$$

Here, the term $\sum_{b=1}^m S(a, b)$ represents the strength of alternative a in outranking the other alternatives (including itself). Conversely, the term $\sum_{b=1}^m S(b, a)$ reflects the extent to which alternative a is outranked by the others (including itself). Hence, R_a reflects the overall strength of alternative a in these outranking relations. A larger R_a indicates that the alternative a is strongly preferred in pairwise comparisons, and hence more desirable. Besides, R_a can be negative, which means that the alternative a is more frequently outranked by others than it outranks them, indicating a relatively weaker overall standing in the decision context. The alternative with the largest outranking score, R_a is top-ranked and recommended to the decision-maker.

Numerical Calculations:

For instance, the outranking score for alternative, $a = 1$, using $S(a, b)$ values in Table 9.5, is:

$$R_1 = \sum_{b=1}^m S(1, b) - \sum_{b=1}^m S(b, 1) = (1 + 0 + 0.4091) - (1 + 0.0288 + 0) = 0.3803$$

Similarly, the outranking scores of all alternatives are calculated as $R_1 = 0.3803$, $R_2 = 0.0288$, and $R_3 = -0.4091$. Accordingly, the ranking is $A1 > A2 > A3$, with A1 being the top-ranked alternative by ELECTRE III.

The advantages of the ELECTRE family are as follows. (1) By relying on outranking relations instead of additive aggregation, ELECTRE methods can express non-compensatory preferences; so, a very poor score for one criterion cannot be fully offset by excellence on others. (2) The method naturally accommodates qualitative, quantitative, and partially ordered data, because concordance–discordance checks are based only on pairwise comparisons rather than explicit utility scales. (3) Several well-defined ELECTRE variants (I, II, III, IV, IS, V, TRI, etc.) can be picked and used by decision-makers, following a familiar conceptual framework. (4) No normalization of the original ACM is required because the method relies on

criterion-specific thresholds (indifference, preference, and veto), which inherently account for differences in measurement scales.

The limitations of the ELECTRE family are as follows. (1) Besides weights, most variants require additional inputs such as preference thresholds for every criterion, which can be cognitively demanding to elicit and may increase subjectivity. (2) The use of veto and discordance in ELECTRE methods can result in some alternatives being incomparable (i.e., no clear preference can be established between them due to conflicting evidence such as strong concordance support overridden by high discordance or veto). While this reflects real-world uncertainty and can be useful, too many such instances of incomparability may frustrate decision-makers, who typically prefer a complete ranking of alternatives. (3) With an increasing number of alternatives and/or criteria, the comparison matrices (i.e., concordance, discordance, and credibility) grows quickly, making ELECTRE methods computationally intensive.

9.3 Preference Ranking Organization Method for Enrichment Evaluations (PROMETHEE) Methods

The PROMETHEE family, first put forward by Brans et al. (1982), belongs to the same class of outranking-type as ELECTRE methods. It relies on pairwise comparisons of alternatives on every criterion, converting each performance difference into a preference degree through one of six generalized preference functions outlined in Brans et al. (1986), and then aggregating those degrees (usually with simple additive weights) into multi-criteria preference indices. From these indices, it then computes the leaving and entering flows that express how much each alternative outranks or is outranked by the others.

Like the ELECTRE family, PROMETHEE family also has several variants, such as PROMETHEE I, II, III, IV, V, and VI (Behzadian et al., 2010). PROMETHEE I (Brans et al., 1982) provides a partial ranking of alternatives based on preference flows, allowing for incomparability between some options. PROMETHEE II (Brans et al., 1986) extends this by delivering a complete ranking using net preference flows, making all alternatives directly comparable (Alinezhad & Khalili, 2019). PROMETHEE III introduces interval evaluations to incorporate uncertainty, considering alternatives as indifferent if their preference flows are very close (Brans & De Smet, 2005; Fernández-Castro & Jiménez, 2005).

The next variant, PROMETHEE IV is adapted for problems involving a continuous set of alternatives, often using integral calculus to evaluate preference distributions (Brans & De Smet, 2005; Albuquerque & Montenegro, 2016). PROMETHEE V (Brans & Mareschal, 1992) incorporates optimization under constraints by integrating integer linear programming, which is useful for selecting the best subset of actions under limited resources (Fernández-Castro &

Jiménez, 2005). PROMETHEE VI (Brans & Mareschal, 1995) handles imprecise criteria weights by defining them as intervals instead of fixed values; it calculates the stability range of possible net flows for each alternative, visualizing the results in a profile called “human brain” (Figueira et al., 2005).

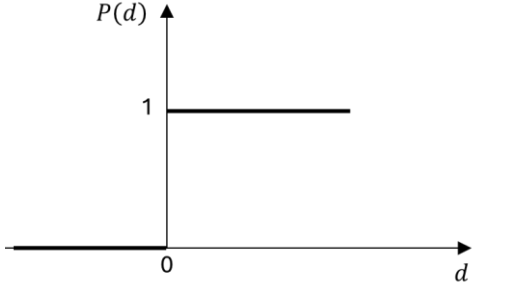
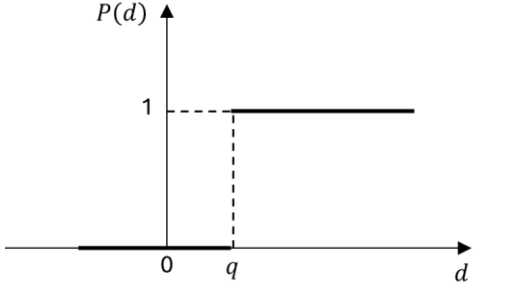
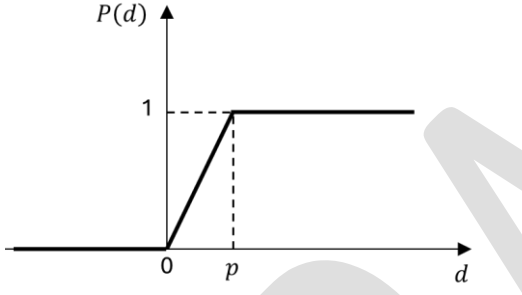
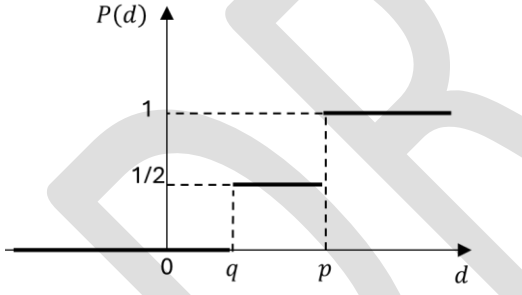
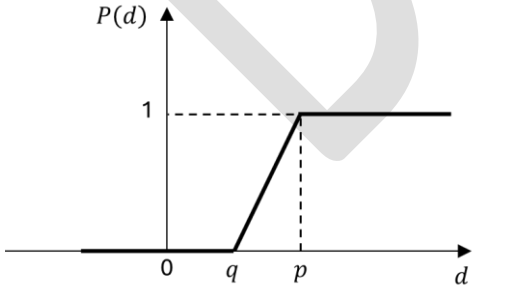
Among these variants, PROMETHEE II is consistently reported as the most widely applied method of the family (Torbacki, 2021; Taherdoost & Madanchian, 2023), because it is computationally straightforward and conceptually simple, as well as it is the first method in the PROMETHEE family, which can generate a complete ranking of alternatives. Hence, the steps of PROMETHEE II are outlined below.

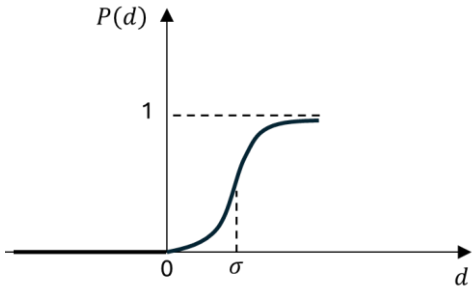
Step 1. Choose a generalized preference function $P_j(d)$ from Table 9.6 for every criterion j in the ACM. The preference function need not be the same for all criteria. In this table, d represents the difference between the two alternatives (a and b) with respect to the criterion j . In this step, the indifference threshold q (non-negative) and/or the preference threshold p (non-negative) must be specified by the decision-maker. The indifference threshold q sets the largest gap on a criterion that the decision-maker still regards as negligible (i.e., when the difference in the values of a criterion for two alternatives is less than or equal to q , the two alternatives are treated as essentially equal). The preference threshold p marks the point where the gap becomes clearly significant, giving full or strict preference to the better performer.

The general guidance on choosing a preference function for a criterion is as follows. Type I is used when any non-zero difference implies a strict preference, making it suitable for criteria where even the slightest improvement is meaningful. Type II is appropriate when only differences beyond a certain threshold q are considered significant, meaning that there is an indifference zone. Type III fits situations where preference increases linearly with the difference from zero up to a predefined threshold p , ideal for criteria with proportional sensitivity. Type IV introduces three zones: indifference up to q , a middle range of weak preference between q and p , and strict preference beyond p , making it useful when gradations of importance are recognized. Type V further refines Type IV by assuming small differences up to q are negligible, but preferences increase linearly between q and p . Lastly, Type VI is best suited for continuous, smooth data where the preference increases gradually, commonly applied in criteria requiring sensitivity to nuanced differences.

Table 9.6: Generalized preference functions used in PROMETHEE methods (Brans et al., 1986)

Generalized preference function	Definition	Parameters
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	Type I: Usual criterion $P(d) = \begin{cases} 0 & \text{if } d \leq 0 \\ 1 & \text{if } d > 0 \end{cases}$	Nil
	Type II: Quasi-criterion $P(d) = \begin{cases} 0 & \text{if } d \leq q \\ 1 & \text{if } d > q \end{cases}$	q
	Type III: Criterion with linear preference $P(d) = \begin{cases} 0 & \text{if } d \leq 0 \\ \frac{d}{p} & \text{if } 0 < d \leq p \\ 1 & \text{if } d > p \end{cases}$	p
	Type IV: Level criterion $P(d) = \begin{cases} 0 & \text{if } d \leq q \\ \frac{1}{2} & \text{if } q < d \leq p \\ 1 & \text{if } d > p \end{cases}$	q, p
	Type V: Criterion with linear preference and indifference area $P(d) = \begin{cases} 0 & \text{if } d \leq q \\ \frac{d - q}{p - q} & \text{if } q < d \leq p \\ 1 & \text{if } d > p \end{cases}$	q, p

	Type VI: Gaussian criterion $P(d) = \begin{cases} 0 & \text{if } d \leq 0 \\ 1 - e^{\frac{-d^2}{2\sigma^2}} & \text{if } d > 0 \end{cases}$	σ (standard deviation)
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Numerical Calculations (using ACM from Table 9.1):

For instance, we have selected the preference functions for the 4 criteria and specified the corresponding parameter values, as presented in Table 9.7. These selections and values are for demonstration, and they can be changed.

Table 9.7: ACM with preference functions and parameter values for PROMETHEE II walkthrough.

Alternatives	C1	C2	C3	C4
A1	0.78	117	6.2	-35
A2	0.65	150	5.8	-40
A3	0.83	110	4.9	-30
Preference function	Type I	Type V	Type III	Type II
q	-	5	-	8
p	-	10	0.8	-

Step 2. Compute preference degree for each pair of alternatives (a and b) under criterion j .

$$d_{ab}^{(j)} = \begin{cases} f_j(a) - f_j(b) & \text{if criterion } j \text{ is to maximize} \\ f_j(b) - f_j(a) & \text{if criterion } j \text{ is to minimize} \end{cases} \quad (9.6a)$$

As such, positive $d_{ab}^{(j)}$ always favors alternative a . Next, $d_{ab}^{(j)}$ is fed into the chosen $P_j(d)$ function to compute the preference degree of alternatives a and b under criterion j as follows.

$$\text{Preference degree} = P_j(d_{ab}^{(j)}) \quad (9.6b)$$

Overall, $P_j(d_{ab}^{(j)}) = 0$ signifies an indifference between a and b under criterion j . In contrast, $P_j(d_{ab}^{(j)}) = 1$ means strict preference of a over b with respect to criterion. A special case is when an alternative is compared to itself (i.e., $a = b$), it always holds that $P_j(d_{ab}^{(j)}) = 0$.

Numerical Calculations:

For instance, consider criterion $j = 3$ (maximization/benefit criterion) and compare the two alternatives of $a = 1$ and $b = 2$. The performance gap is:

$$d_{12}^{(3)} = f_3(1) - f_3(2) = 6.2 - 5.8 = 0.4$$

As given in Table 9.7, this criterion employs Type III with a preference threshold $p = 0.8$. Because $0 < d_{12}^{(3)} \leq p$, using the Type III preference definition from Table 9.6, the preference degree is obtained:

$$P_3(d_{12}^{(3)}) = \frac{d_{12}^{(3)}}{p} = \frac{0.4}{0.8} = 0.5$$

Thus, alternative $a = 1$ is accorded with a moderate preference of 0.5 over alternative $b = 2$ on criterion 3.

Now, for the same benefit criterion $j = 3$, let us compare the two alternatives the other way around, i.e., $a = 2$ and $b = 1$. The performance gap is:

$$d_{21}^{(3)} = f_3(2) - f_3(1) = 5.8 - 6.2 = -0.4$$

This criterion employs Type III and $d_{21}^{(3)} \leq 0$. Using the Type III preference definition from Table 9.6, the preference degree is obtained:

$$P_3(d_{21}^{(3)}) = 0$$

Similarly, following this comparison and calculation using the chosen preference function, the preference degree for other pairs of alternatives under each criterion can be obtained. The complete preference degree matrices are presented in Table 9.8. As observed, each preference degree matrix is not symmetric along the principal diagonal, since $P_j(d_{ab}^{(j)})$ does not necessarily equal $P_j(d_{ba}^{(j)})$.

Table 9.8: Preference degree matrices for PROMETHEE II walkthrough.

$P_1(d)$	A1	A2	A3
A1	0	1	0
A2	0	0	0
A3	1	1	0
$P_2(d)$	A1	A2	A3
A1	0	0	0.4
A2	1	0	1
A3	0	0	0
$P_3(d)$	A1	A2	A3
A1	0	0.5	1

A2	0	0	1
A3	0	0	0
$P_4(d)$	A1	A2	A3
A1	0	0	0
A2	0	0	0
A3	0	1	0

Step 3. Aggregate the preference degrees to build the multi-criteria preference matrix $\pi(a, b)$.

$$\pi(a, b) = \sum_{j=1}^n w_j P_j(d_{ab}^{(j)}) \quad (9.7)$$

Numerical Calculations:

For instance, for the two alternatives of $a = 1$ and $b = 2$:

$$\pi(1, 2) = \sum_{j=1}^4 w_j P_j(d_{12}^{(j)}) = 0.30 \times 1 + 0.25 \times 0 + 0.20 \times 0.5 + 0.25 \times 0 = 0.4$$

Similarly, the remaining elements of the multi-criteria preference matrix can be computed. The complete matrix is presented in Table 9.9. This matrix is not symmetric as the individual preference matrices are not symmetric.

Table 9.9: Multi-criteria preference matrix for PROMETHEE II walkthrough.

Alternatives	A1	A2	A3
A1	0	0.4	0.3
A2	0.25	0	0.45
A3	0.3	0.55	0

Step 4. Calculate the outranking score R_a for each alternative a .

$$R_a = \sum_{b=1}^m \pi(a, b) - \sum_{b=1}^m \pi(b, a) \quad (9.8)$$

Here, the sum, $\sum_{b=1}^m \pi(a, b)$ is the positive flow, measuring the extent to which alternative a outranks all other alternatives (including itself). Conversely, the sum, $\sum_{b=1}^m \pi(b, a)$ is the negative flow, indicating how strongly a is outranked by the other alternatives (including itself). Hence, R_a reflects the overall strength of alternative a in these outranking relations. The alternative with the largest R_a is top-ranked and recommended to the decision-maker.

Numerical Calculations:

For instance, for alternative $a = 1$:

$$R_1 = \sum_{b=1}^m \pi(1, b) - \sum_{b=1}^m \pi(b, 1) = (0 + 0.4 + 0.3) - (0 + 0.25 + 0.3) = 0.15$$

Analogously, outranking scores of all alternatives are calculated as $R_1 = 0.15$, $R_2 = -0.25$, and $R_3 = 0.1$. Accordingly, the ranking is $A1 > A3 > A2$, with A1 being the top-ranked alternative by PROMETHEE II.

The advantages of the PROMETHEE family are as follows. (1) PROMETHEE methods deliver a complete ranking for every set of alternatives, sparing decision-makers the frustration of incomparable and possibly many “ties” as in ELECTRE methods. (2) A coherent suite of variants uses the same underlying preference mechanism, and so decision-makers can switch from one variant of PROMETHEE to another with ease. (3) No normalization of the original ACM is required because the use of preference functions and associated thresholds allows each criterion to be evaluated directly in its own units. (4) PROMETHEE methods are computationally slightly lighter than ELECTRE methods, as its steps are more straightforward and involve fewer calculations.

The limitations of the PROMETHEE family are as follows. (1) Beyond assigning criterion weights, decision-makers must also choose a preference function for each criterion and specify parameters such as the indifference threshold q , preference threshold p or Gaussian spread σ ; eliciting these inputs can be cognitively demanding and inject subjectivity. (2) Because PROMETHEE requires a pairwise comparison for every pair of alternatives, the size of the preference matrix grows rapidly, making it computationally intensive.

9.4 Summary

Main points in this chapter on outranking-type MCDM methods are as follows.

1. This chapter describes two prominent outranking-type MCDM families (ELECTRE and PROMETHEE). These methods evaluate alternatives using pairwise comparisons to establish outranking relationships.
2. The most widely used variant of each family, namely, ELECTRE III and PROMETHEE II, are described and illustrated using a simple numerical example, enabling intuitive understanding of their respective mechanisms.
3. Outranking-type MCDM methods assess the extent to which one alternative outranks another, incorporating notions such as indifference and preference thresholds. These

thresholds allow decision-makers to express nuanced preferences that might not be captured by purely additive and multiplicative approaches.

4. Each method in this chapter is systematically presented with a clear explanation of its theoretical foundation, algorithmic steps, and numerical calculations using a common ACM. This helps readers grasp the practical implementation of each method.
5. The chapter discusses the advantages and limitations associated with ELECTRE and PROMETHEE families. Both methods do not require normalization of ACM. However, they require additional parameters beyond criterion weights, such as threshold values and/or preference functions, which enhance flexibility but also introduce subjectivity and cognitive burden for decision-makers.
6. Finally, using the common ACM presented in Table 9.1, both ELECTRE III and PROMETHEE II identify A1 as the top-ranked alternative. The ranking order of the remaining alternatives differs between the two methods, reflecting their underlying methodological distinctions. These ranking results cannot be generalized, and it is advisable to apply multiple MCDM methods when solving real-world problems.

9.5 References

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9.6 Exercises

- E9.1 You are given an ACM consisting of 3 alternatives (A1, A2, A3) and 4 criteria (C1, C2, C3, C4), with the following values:

Alternatives	C1	C2	C3	C4
A1	1.39	358	0.161	2.857
A2	1.55	448	0.172	2.5
A3	1.88	328	0.204	3.333

Criteria C1 and C2 are benefit criteria to be maximized, whereas C3 and C4 are cost criteria to be minimized. The weights for the criteria are: $w_1 = 0.25$, $w_2 = 0.23$, $w_3 = 0.33$, and $w_4 = 0.19$. Apply both the outranking-type MCDM methods covered in this chapter to rank the alternatives. Identify the top-ranked alternative by each method and compare them.

- E9.2 Briefly explain the role of veto threshold (V_j) in ELECTRE III. Conduct sensitivity analysis by changing their values for the ACM in exercise E9.1. Comment on your findings.
- E9.3 Briefly explain the role of the parameters q and p in PROMETHEE II. Conduct sensitivity analysis by changing their values for the ACM in exercise E9.1. Comment on your findings.
- E9.4 You are given an ACM consisting of 4 alternatives (A1, A2, A3, A4) and 3 criteria (C1, C2, C3), with the following values:

Alternatives	C1	C2	C3
A1	52.9	25.3	0.346
A2	46.6	23.5	0.367
A3	59.8	28.3	0.332
A4	57.8	25.9	0.308

Criteria C1 and C2 are benefit criteria to be maximized, whereas C3 is a cost criterion to be minimized. The weights for the criteria are: $w_1 = 0.35$, $w_2 = 0.39$, and $w_3 = 0.26$. Apply both the outranking-type MCDM methods covered in this chapter to rank the alternatives. Identify the top-ranked alternative by each method and compare them.

- E9.5 Describe the similarities and differences between the ELECTRE III and PROMETHEE II methods.